Preet Patel

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Education

Master of Science in Computer Science | University of Illinois, Chicago **Expected: May 2027**
Coursework: Cloud Computing, Computer Systems Security, Secured Web-Application Development

Bachelor of Technology in Computer Science | Ahmedabad University **May 2025**
Coursework: Artificial Intelligence, Machine Learning, Advanced Business Analytics, Computer Networks

Internship Experience

yourEDUGATOR **May 2025 – August 2025**
Product Development Intern *Ahmedabad, Onsite*

- Designed the IELTS Speaking module using Azure Speech SDK, ElevenLabs Speech-to-Text, and OpenAI Text-to-Speech, improving end-to-end speech evaluation throughput by 3×.
- Collaborated directly with CEO to develop an automated IELTS-Writing Task-1 diagram generator (DALL·E), and JSON schemas for IELTS Reading and Listening used in production to expand question coverage by 120%.
- Developed a ML-based metrics evaluation scoring engine and GenAI feedback system to evaluate student performance; achieved 80% accuracy in predicting band scores across individual modules.

Techsture Technologies Ind. Pvt. Ltd. **May 2024 – August 2024**
Research Intern *Ahmedabad, Onsite*

- Developed a YOLOv4-based real-time pedestrian detection model and deployed it on infrastructure (RSU) and created an alert dissemination pipeline utilising TCP/IP to notify the smart vehicles (OBU) nearby via I2V communication.
- Created a CNN-based driver drowsiness detection model and deployed it on raspberry-powered device with an in-built camera in the car, which constantly monitors the driver's eye movements.
- Built a live tracking dashboard using JavaFX and OpenStreetMap to visualize vehicle locations and created RESTful APIs to get the alerts and vehicle postions.

Technical Skills

- **Languages**: Python, JavaScript/TypeScript, Go, Scala, C/C++, Java, SQL
- **Cloud & DevOps**: AWS (EC2, EMR, S3, EKS, Lambda), Kubernetes, Git, Docker
- **Frameworks**: Tensorflow, Pytorch, sci-kit learn, Hugging Face, Langchain, OpenCV, MediaPipe
- **Distributed Systems/Data**: Apache (Spark, Hadoop, Flink, Lucene, Kafka), Delta Lake, MongoDB

Projects

GraphRAG: Distributed Data Pipeline Deployed on AWS **September 2025 – November 2025**

- Engineered a distributed pipeline using Hadoop MapReduce + Lucene to ingest and index 600+ PDFs, building stable chunk/embedding stores with content-hash deduplication and efficient k-NN search over float-vector fields.
- Implemented an incremental, fault-tolerant layer using Spark + Delta Lake, enabling re-indexing when documents or embedding models changed, and reducing recomputation by 90% through deterministic delta detection and metadata.
- Designed a cloud-native GraphRAG system on AWS using Flink, Neo4j, and Ollama inference, streaming chunks through relation extraction, LLM scoring, and idempotent graph upserts inside a Flink-on-EKS deployment.

GO-Emotion: Emotion Classifier using BiLSTM + GloVe **October 2025 – November 2025**

- Developed a production-ready Flask inference service for multi-label emotion classification, leveraging TensorFlow BiLSTM models with GloVe embeddings to serve low-latency NLP predictions at scale.
- Designed a custom focal-loss objective to address extreme label imbalance across 28 emotion classes, improving hard-example learning and stabilizing model convergence during training.
- Built an end-to-end ML serving pipeline (tokenization, embedding lookup, model inference, JSON API response), enabling seamless integration of deep-learning models into downstream applications.

Publications

P.Patel and D.K.Patel, "**Capacity Analysis of Vehicular Networks in a Multi-User Mixed Traffic Scenario**," NCC2025, IEEE, pp.1-6. DOI:10.1109/NCC63735.2025.10983153

Leadership & Activities

- **Research Scholar, MICxN Lab:** Led 5+ workshops and reproduced 15+ research papers in the tenure of 1.5 years. Reviewed 50+ manuscripts of the junior research scholars.
- **Clash of Codes:** Team ranked 1/20 in a competitive coding-based team competition organised by the university in season 1 (2023) and 2/20 in season 2 (2024); 4/80 in individual scores and 3rd in problems solved.
- **Teaching Assistant & Peer Tutor, Ahmedabad University:** Assisted the professors in courses like AI, Embedded Systems, Computer Networks, and Predictive Analytics for Business (classes up to 130 students).